

Projects and contact details

The Soil Science Challenge Grants is a \$20 million (GST Excl.) competitive grants program that supports universities to research critical gaps in soil science in one of the following areas: soil hydrology, soil carbon dynamics, soil nutrient uptake, and soil/root interface.

In 2022, eleven Grant Agreements were executed with multiple universities from 2021–22 to 2024–25.

Vertosols cropping project (The University of Queensland) **(\$1.59 million GST Excl.)**

Cropping farmers spend more than \$1 billion in farming inputs (nitrogen). The efficiency of this input on those crops is often less than 25 per cent. This project will refine nitrogen fertiliser strategies for a significant portion of cropping producers in Australia. This is an important area of research to address the challenges to Australia's ongoing agricultural productivity potential. More efficient and effective strategies to the use of nitrogen will be a key part of Australia's adaptation to climate change by also providing additional information about water dynamics.

Project Lead: [Professor Michael Bell](#)

Phosphorous project (The University of Queensland) **(\$0.52 million GST Excl.)**

Two thirds of global soils are deficient in phosphorous and long-term cropping is reducing these levels even further. This project aims to unravel changes in the concentration, availability, and speciation of phosphorous in agricultural soils after use for up to 100 years. The project will provide a better understanding of phosphorous reserves and its application in soils to increase agricultural productivity.

Project Lead: [Professor Peter Kopittke](#)

Soil fauna in agricultural soils (The University of Melbourne) **(\$1.97 million GST Excl.)**

Soil fauna is important in promoting the productivity of all soil types. This project will provide the first real insight into how we manage soil fauna with a view to influencing agricultural productivity across a broad range of agricultural industries. This project will provide farmers and land managers with the necessary tools to assess soil fauna in their soil to enable them to better influence soil productivity by developing new standardised methodologies for management.

Project Lead: [Professor Jim He](#)

Quantification, causes and consequences of soil organic carbon decline in Australian cropping systems (The University of Melbourne)

(\$1.78 million GST Excl.)

Australia is a global leader in crop productivity and nutrient efficiency which is threatened by removing nitrogen and carbon in our soil without the appropriate input in our extensive cropping landscapes. This project aims to ensure the appropriate balance of inputs to sustain crops. A major constraint to agricultural productivity is the fixed supply of nitrogen. This project aims to maximise the productivity and security of nitrogen, one of the main inputs in agriculture. This project will also support our understanding of carbon sequestration because an ability to balance nitrogen in crop demand is critical to enable carbon sequestration to occur.

Project Lead: [Professor Deli Chen](#)

Soil carbon project (The University of Sydney)

(\$0.77 million GST Excl.)

There is current significant interest in the potential of soil carbon to support a response to a changing climate. An important part of that potential is in soil carbon sequestration. This project investigates one of the major candidates for carbon sequestration in soil, provides an understanding of how it works, and what the mechanism is (including if it is stable). This project will look at the interaction of various components in soil to understand the actual sequestration linkages and their benefits.

Project Lead: [Professor Balwant Singh](#)

Soil viral diversity (The University of Sydney)

(\$2.69 million GST Excl.)

This project will deliver globally innovative work into the diversity and function of viruses in soils across the Australian landscape. Currently, there is limited knowledge about the impact of viral health on soil and soil function. Viruses are the most abundant organisms on the planet. This project is the opportunity for Australia to be the first nation in the world to examine the nature of viruses in soil. The project has a significant opportunity to benefit soil health and productivity in Australia and globally.

Project Lead: [Professor Budiman Minasny](#)

High resolution monitoring (The University of Sydney)

(\$1.67 million GST Excl.)

As climate and climate management have changed substantially over time, our ability to monitor those changes need to be developed. There are new ways of organising soil knowledge that can have a fundamental impact on our understanding of the distribution of soils in the landscape and how that distribution can impact on farmers and primary producers. How we understand soil types and distribution, and how we map this distribution can also have a significant impact on our ability to

predict things more accurately such as soil function and locations that might be susceptible to erosion. This project will develop an improved spatial understanding of how soils behave in the landscape, to enable us to develop better strategies for managing necessary responses.

Project Lead: [Professor Budiman Minasny](#)

Past, present, and future drivers of soil change (The University of Adelaide) (\$2.84 million GST Excl.)

Building on a unique archive of soil and DNA information, this project is innovative and globally unique. It aims to understand diversity in soil from a new approach utilising DNA technology in soil samples. This project is equivalent to the full genomic sequencing of 10,000 soil samples.

Project Lead: [Associate Professor Stuart Roy](#)

Soil biological mechanisms (The University of Western Australia) (\$3.94 million GST Excl.)

This project will improve the ability to understand the use of organic materials and their specific benefit to inform the development of strategies to capitalise on organic additions to our soils and maximise productivity and output benefits. The generation of organic materials is crucial to the sustainability of our systems, and microbial processes in soil, and this project will fill a critical research gap in this area.

Project Lead: [Dr Sasha Jenkins](#)

Engineering novel carbon amendments (The University of NSW) (\$1.10 million GST Excl.)

The use of nano materials has revolutionised a whole range of fields that affect a broad area of health and environmental outcomes. This project trials the use of nano materials, often including waste materials, to assist in carbon sequestration.

Project Lead: [Professor Paul Monroe](#)

Grasslands Project (The Queensland University of Technology) (\$0.80 million GST Excl.)

The northern Australia pastoral zone is a significant contributor to Australia's national exports. This zone also possesses significant potential to contribute to Australia's carbon storage. Improved strategies to better manage these vast areas of pastoral land has a substantial potential to improve both carbon storage and productivity on that land. This project will provide a fundamental understanding that will lead to enhanced rangelands management. Given the large scale of the area involved, even small gains in an approach over such a large geographic area has the potential for realising significant benefit, in an area that has previously been overlooked.

Project Lead: [Associate Professor David Rowlings](#)

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/natural-resources/soils/science-challenge-grants>